

What is Parquet?

Apache Parquet is a columnar storage file format used for storing large datasets efficiently.

CSV → stores data row by row

Parquet → stores data column by column

This makes Parquet:

Faster for analytics

Smaller file size (compressed)

Supported by Python, Java, Spark, Hadoop, etc.

Read Parquet using Pandas

```
import pandas as pd
```

```
df = pd.read_parquet("employees.parquet")
```

```
print(df.head())
```

Write Parquet using Pandas

```
import pandas as pd
```

```
data = {  
    "name": ["John", "Alice"],  
    "age": [25, 30]  
}
```

```
df = pd.DataFrame(data)
```

```
df.to_parquet("employees.parquet")
```

Required Libraries

Install one of these engines:

```
bash
```

```
pip install pyarrow
```

or

```
bash
pip install fastparquet
```

Using PyArrow Directly

```
import pyarrow.parquet as pq

table = pq.read_table("employees.parquet")
df = table.to_pandas()

print(df)
```

CSV vs Parquet

Feature	CSV	Parquet
Human readable	Yes	No
Compression	No	Yes
Storage Size	Large	Small

Read Speed	Slower	Faster	
Column-based	No	Yes	
Spark Support	Yes	Excellent	

Can Java Read Parquet?

Yes. Parquet is not a Python-specific format. It can be read by:

Java

Apache Spark

Hadoop

Python (Pandas, PyArrow)

Databricks

Convert CSV to Parquet

```
import pandas as pd
```

```
df = pd.read_csv("employees.csv")  
df.to_parquet("employees.parquet")
```

So, Parquet is a file format, not a Python library. In Python, you typically use Pandas + PyArrow to work with Parquet files.

